

QBI HACKATHON 2026 · UCSF MISSION BAY · MIT-LICENSED

CryoClear

The open, real-time particle-picking +
junk-triage copilot for cryo-EM.

CryoFSL made few-shot picking accurate. **We make picking and junk-cleaning interactive, real-time, and measurable** — with a human-in-the-loop that learns live, on a single GPU.

Computer Science · picking + junk ML + UI

Hardware / Systems · GPU, MRC I/O, streaming

Chemistry · scientific story + HITL

THE GAP

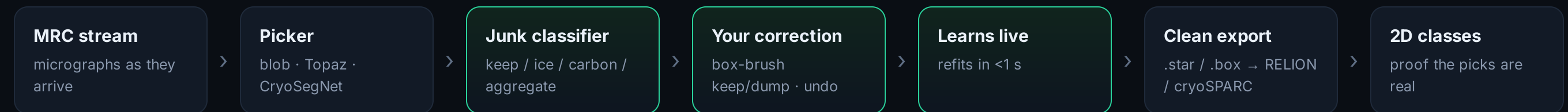
Auto-pickers over-pick junk. Cleaning it is manual, post-hoc, and slow.

- Topaz, crYOLO, CryoSegNet find particles well — but ship no junk-triage layer for ice, carbon edges, and aggregates.
- Scientists catch junk by eye, after the run — discard, re-tune, re-run. Hours after collection, not during.
- Real-time platforms (cryoSPARC Live, Warp) use simple pickers and are heavy / closed.

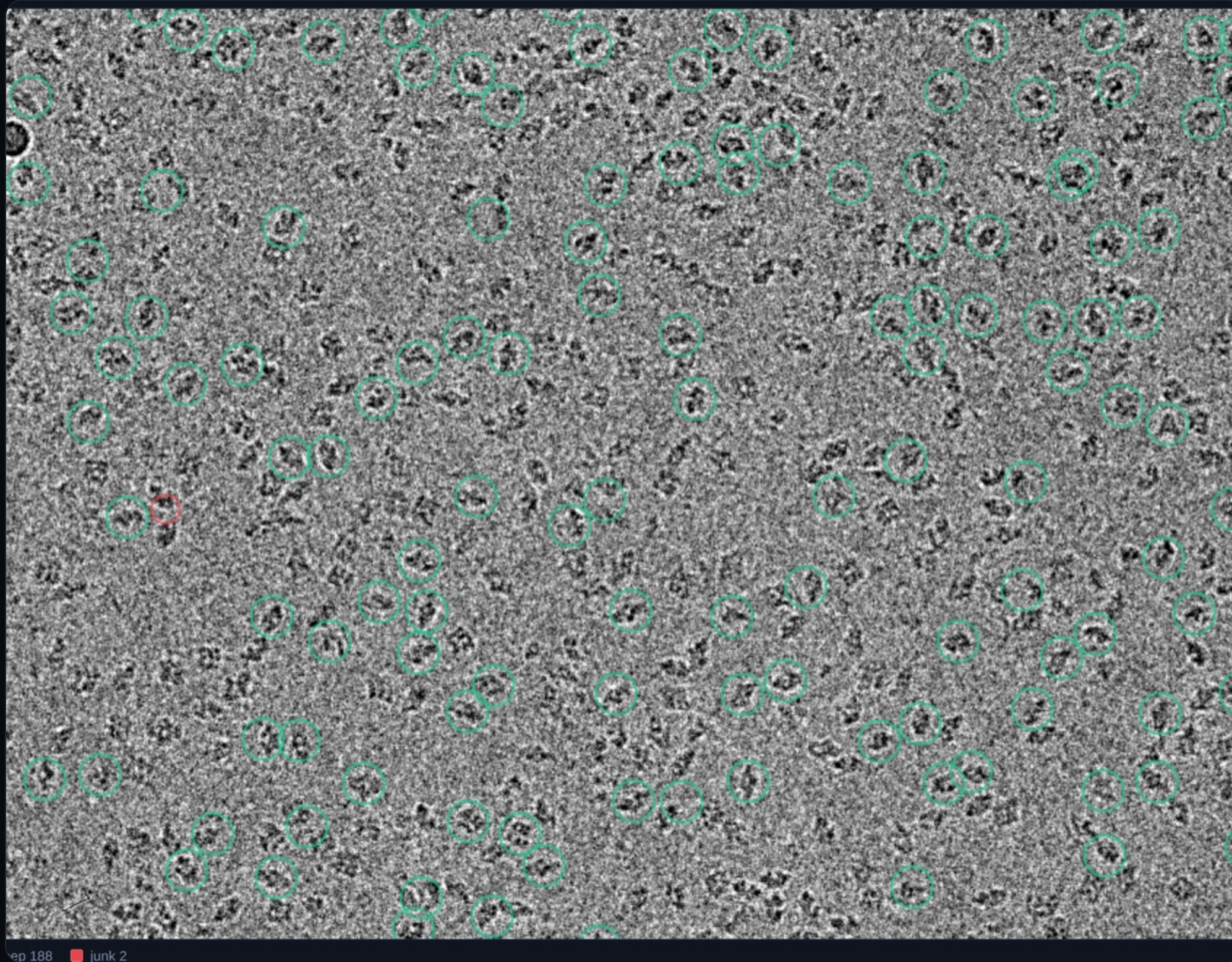
The opening: nobody ships an **open, interactive, real-time junk-triage copilot** with a live human-in-the-loop learning loop. That product — not a new picking model — is our contribution.

WHAT CRYOCLEAR DOES

One modular pipeline — swap any picker, triage the junk, learn from the scientist.



Green = keep, red = junk, overlaid live. Every accept/reject updates the classifier in-session — and a live scoreboard shows exactly how clean the stack is.



ep 188 ■ junk 2

EMPIAR-10017 β -galactosidase · each **green** ring is one picked particle, **red** is flagged junk. Pan/zoom is $O(1)$ on an off-screen canvas.

**It picks
particles —
and tells you
which ones
to trust.**

- Locally-normalised band-pass detector finds candidates across the whole field.
- Each candidate gets 23 intensity-normalised features (radial profile, edge coherence, sharpness).
- The classifier paints keep vs junk in real time.

MEASURED, NOT ASSERTED

Junk triage beats raw picks where it counts: a cleaner stack.

+0.09

kept-set purity vs raw picks
improves on every micrograph (in-sample)

27%

pure junk removed
recall unchanged — no real particles lost

0.36 → 0.49

precision at the clean operating point

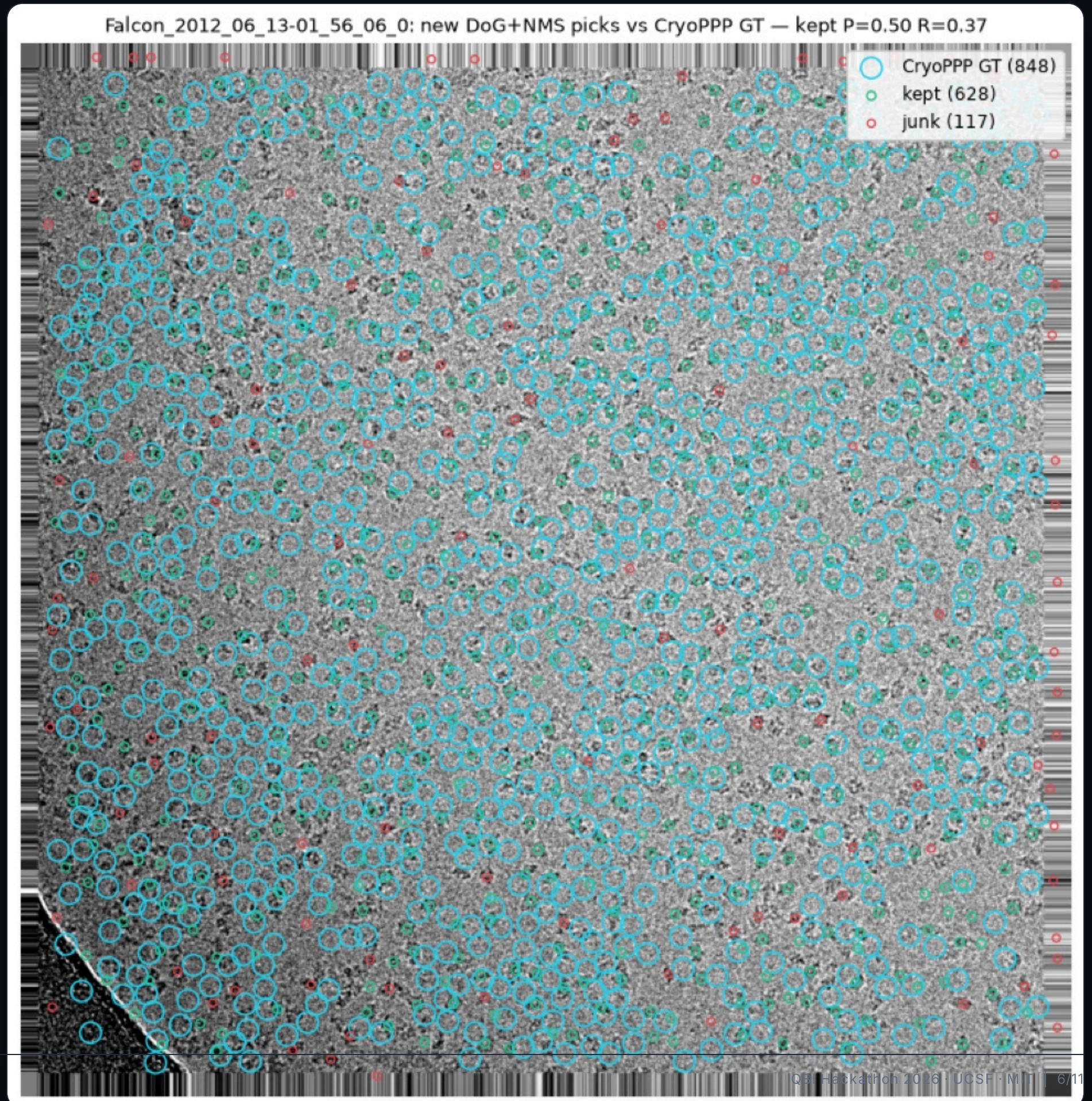
~0.6–0.75

junk-rejection F1 (held-out)

Honest ceiling: on β -gal, picking-F1 after triage $\approx$ raw — the leftover false positives are background that genuinely *resembles* particles, which no classifier (cryoSPARC's included) can separate. The model's real, measurable value is **purity and a clean kept set**, not a picking-F1 lift.

Scored against expert CryoPPP ground truth.

- Picks matched to CryoPPP's hand-labelled particles within the particle radius.
- We report precision / recall / F1 per micrograph — and held-out, not just in-sample.
- Published deep-learning pickers reach ~0.73 F1 on their own protocols; we show that for context, not a head-to-head — the picker is a swappable default, the triage layer is the contribution.



It learns from your corrections — live.

- Cold-start: the model begins junk-blind.
- Box-brush to keep or dump particles in bulk; the junk-rejection F1 climbs as it learns.
- Full undo / redo (Ctrl/Cmd+Z); every change flows into the export.

This is the "it learns from me" loop that batch CLI pickers don't have.

Flags each candidate as keep or junk (carbon / ice / aggregate) and removes it from the kept set, lifting purity while keeping the real particles. LightGBM is the robust default; RandomForest, SGD (online), and the CNN are alternatives. Scoreboard is in-sample on these micrographs; held-out generalisation is more conservative.

LEARNING MODE

Trained model

Active-learning

Reset

Cold-start: the model starts junk-blind. Dump/keep particles and the junk-rejection F1 climbs as it learns.



REAL-TIME STREAM

Start stream

EXPORT

.star

.box

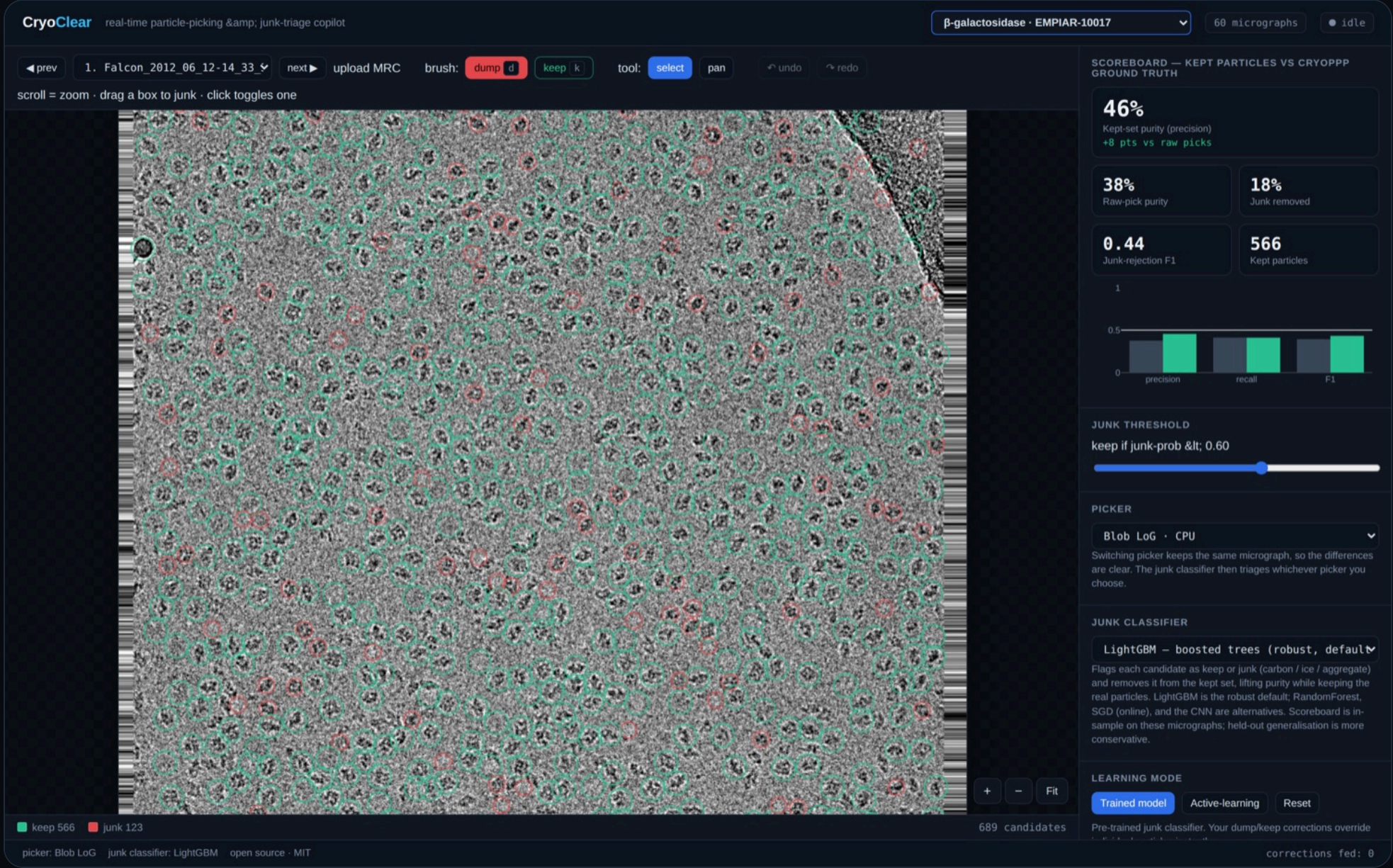
PNG

JPEG

PDF report

.star / .box = kept-particle coordinates for RELION & cryoSPARC; PNG/JPEG = overlay snapshot; PDF = honest scorecard

A PLATFORM, NOT A ONE-OFF



The live app: canvas viewer, scoreboard, swappable picker + classifier, multi-dataset switcher.

Everything is
swappable
and cached.

PICKERS

- Blob LoG · CPU
- Topaz · GPU
- CryoSegNet · GPU

JUNK CLASSIFIERS

- LightGBM
- RandomForest
- SGD · online
- CNN

DATASETS

- β-gal
- TMV
- proteasome
- + upload your own MRC

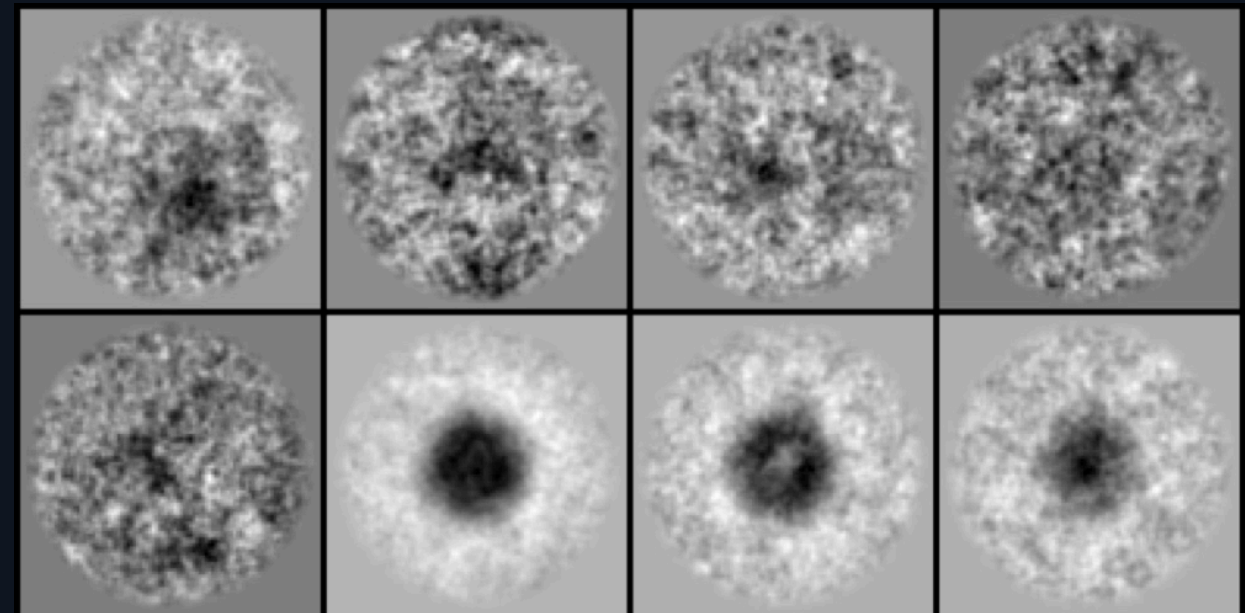
EXPORT

- .star
- .box
- PNG
- PDF report

2D class averages — "these picks are real."

- Reference-free 2D classification of the kept particles.
- cryoSPARC-style expectation-maximisation: translation + rotation alignment, phase-correlation, resolution annealing.
- Recognisable, centred protein views emerge from the cleaned stack.

Honest: β -gal is small, low-contrast, few-view — averages won't match cryoSPARC's textbook proteasome crispness without thousands of particles + CTF. The pipeline is industry-shaped; the protein is the limit.



Class-average montage of kept β -gal particles.

WHY IT SCORES

The novelty is the product — and we're honest about what it isn't.

WE BUILT

- An open, real-time junk-triage copilot with a live HITL learning loop.
- Swappable pickers + classifiers, 7+ datasets, upload-your-own, clean export.
- Honest measurement against CryoPPP — held-out, in-sample labelled as such.

Rubric fit: Impact (cleaner stacks, live) · Execution (a usable product, not a notebook) · Background variety (CS + hardware + chemistry) · Novelty (the interactive real-time junk-triage UX nobody ships).

WE DO NOT CLAIM

- A novel picking model or a novel ML algorithm.
- To beat deep-learning pickers' picking-F1 — swap one in; it's in the menu.
- cryoSPARC-grade 2D on a hard protein from a laptop.

OPEN SOURCE · MIT · SINGLE GPU

Picking + junk-cleaning, interactive, real-time, and measurable.

CryoClear turns post-hoc, manual junk removal into a live copilot that cleans the stack as data streams in — and proves it with honest metrics and 2D class averages.

● Runs on one RTX-class GPU

● MIT-licensed, fully open

● CS · Hardware · Chemistry